19/11/25

Machine Learning
Assignment - 5

Q1

1) Decision trees offer key advantages such as interpretability - their structure is intuitive and easy to visualize.

They handle both categorical and numerical data, function well with missing values, and are robust to outliers.

Decision Trees do not rely on assumptions about the underlying data distribution, so they can be applied to many types of problems.

Disadvantages:

- Overfitting: Trees can become overly complex, fitting minor noise ~~out~~ of the training data, resulting in poor generalization to the unseen data.

- Instability: Small changes in the training data can result in a completely different tree structure, making results unpredictable and non-reliable.

- **Bias toward Dominant Classes:** In cases of imbalanced data, trees may favour the most frequent class and ignore minority classes, reducing predictive power.

- **Computational Expense and Scalability:** For large datasets with many features, training decision trees can be computationally demanding, and the search space for splits increases exponentially.

- **Greedy Nature:** Decision trees build using greedy algorithms optimizing for the best local split, which may not lead to a globally optimal tree.

2) **Some important hyperparameters:**

- **max_depth:** Controls the maximum depth of the tree, limiting how many sequential splits can occur. Smaller values prevent overfitting but may underfit if too restrictive.

- **min_samples_split:** The minimum ~~split~~ number of samples required to split an internal node. Higher values prevent the tree from learning overly specific patterns and reduce overfitting.

- min_samples_leaf : The minimum no. of samples req. to be at a leaf node. This smooths the model by ensuring predictions are made based on sufficient data points.

- max_features : The number of features to consider when looking for the best split. Reducing this adds randomness and can prevent overfitting, particularly useful in ensemble methods.

- cp (complexity parameter) : The cost-complexity pruning parameter that penalizes tree complexity. Higher values lead to more aggressive pruning and simpler trees.

- criterion : The function to measure split quality, such as Gini impurity or entropy for classification, and MSE or MAE for regression.

3) Overfitting a decision tree :

A decision tree overfits when its complexity allows it to model not just underlying trends, but also, noise specific to the training data, leading to poor generalization on new data. Three ways to overfit are :

a) Allow unlimited tree depth: setting 'max-depth' to None or a very high value enables the tree to keep splitting until all leaves are pure causing it to memorize the training data instead of learning generalizable patterns.

b) Very small min-samples_split/min-samples-leaf: Allowing splits with very few samples creates leaves that focus on ~~minute~~ outliers in the data, capturing noise.

c) Fail to apply pruning or regularization: Not employing tree pruning or constraints leaves the tree unchecked, exaggerating complexity ad vulnerability to noise in the training set.